It has to be observed that the maximum purchase falls on the footwear of size 8 inches. So the shopkeeper has to get more number of footwear of size 8 inches. Hence arithmetic mean does not suit for this purpose. Here we need another type of representative value of data called 'Mode'.

Mode is the value of the data which occurs maximum number of times.

Y2R6K1

Consider another example.

A shopkeeper analyses his sales data of readymade shirts to plan for the stock according to the demand. The sale details of shirts are given below.

Size	28"	30"	32"	34"	36"
Number of shirts sold	20	35	22	35	15

Here he observes that there is a equal demand for shirts of sizes 30" and 34". Now this data has two modes as there are two maximum occurrences namely 30" and 34". He stocks more shirts of these 2 sizes. Note that, this data has two mode and it is known as **bimodal data**.

Try these

(1) Find the mode of the following data.

2, 6, 5, 3, 0, 3, 4, 3, 2, 4, 5, 2

(2) Find the mode of the following data set.

3, 12, 15, 3, 4, 12, 11, 3, 12, 9, 19

(3) Find the mode of even numbers within 20.

Example 5.5 Find the mode of the given set of numbers. 5, 7, 10, 12, 4, 5, 3, 10, 3, 4, 5, 7, 9, 10, 5, 12, 16, 20, 5

Solution

Arranging the numbers in ascending order without leaving any value, we get,

3, 3, 4, 4, 5, 5, 5, 5, 5, 7, 7, 9, 10, 10, 10, 12, 12, 16, 20

Mode of this data is 5, because it occurs more number of times than the other values.

Note

To find mode, arranging the raw data in ascending order is not mandatory. It helps us to ensure that each and every value is taken into account for the calculation of mode and helps in identifying the mode value easily.

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